

Gisma University of Applied Sciences

B201 Computer Science Lab

Portfolio Project Report

Assessment Submission Form

Student Number

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(If this is group work, please include the student numbers of all group participants)

Assessment Title

Portfolio Project Report

Module Code

B201D

Module Title

Computer Science Lab

Module Tutor

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Declaration of Authorship

I declare that all material in this assessment is my own work except where there is clear acknowledgement and appropriate reference to the work of others.

I fully understand that the unacknowledged inclusion of another person's writings or ideas or works in this work may be considered plagiarism and that, should a formal investigation process confirm the allegation, I would be subject to the penalties associated with plagiarism, as per Gisma Business School, University of Applied Sciences' regulations for academic misconduct.

Signed. Parvin Riyah Date. 26.06.2026

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1. Introduction

This report details the creation, structural design, and technical workflow of my personal computer science portfolio, developed as a core component of the B201 Computer Science Lab module at Gisma University of Applied Sciences. The goal of this portfolio is to build a cohesive, recognisable professional identity that showcases my software engineering skills and academic progress to prospective employers.

The portfolio ecosystem is built around three pillars: a custom-designed personal website, a structured GitHub repository hosting the code, and a clean, high-fidelity CV typeset in LaTeX. Together, these assets provide a transparent look into my technical proficiencies and development practices.

Links

- **LinkedIn:** <https://www.linkedin.com/in/parvin-riyah-13724a39a>
- **GitHub:** <https://github.com/ParvinRiyah757>
- **GitHub Repostary:** https://github.com/ParvinRiyah757/Parvin-Riyah_portfolio.git
- **Website:** https://github.com/ParvinRiyah757/Parvin-Riyah_portfolio.git
- **CV:** https://github.com/ParvinRiyah757/Parvin-Riyah_portfolio.git
- **Report:** https://github.com/ParvinRiyah757/Parvin-Riyah_portfolio.git

2. Portfolio Website Structure

The portfolio website is engineered as a seamless single-page experience, built from scratch using clean HTML and CSS. It is divided into the following intuitive sections:

- **Hero / Header:** Introduces my name and profile as a computer science student, immediately establishing the site's personal branding.
- **About Me:** A brief overview detailing my background, core values like pragmatism and perfectionism, and my current path.
- **Technical Skills:** A visually striking layout showcasing my proficiencies in Python, HTML/CSS, and SQL/MySQL.
- **Language Metrics:** A dedicated section displaying my language proficiencies, highlighted by my C1 English level, using custom geometric indicators.
- **Project Cards:** Deep dives into my development projects, complete with architecture summaries and direct links to their source code repositories.
- **Education:** A timeline tracking my academic steps, including secondary education and my current Bachelor of Computer Science degree at Gisma University of Applied Sciences.
- **Footer:** Clean, accessible links for direct contact and professional networking.

The page also features an easily accessible download button inside the header/about section, allowing visitors to instantly grab the PDF version of my LaTeX CV.

3. GitHub Repository Structure

To maintain an organized development workspace, the portfolio repository is structured cleanly, separating core frontend code from underlying project assets:

```
parvin-riyah-portfolio/  
|-- Report      # Compiled LaTeX project report  
|-- Cv          # Compiled LaTeX CV for direct download  
+-- Website     #Main HTML5 structure and layout rules
```

While smaller files are stored directly within the primary portfolio repository, larger database components—such as my E-sports Database Management System—are hosted in their own dedicated GitHub repositories to preserve distinct version control histories and separate documentation.

4. Design Decisions

4.1 Website Design & The Blue Aesthetic

I chose to build this website using pure HTML5 and CSS3 without touching heavy frameworks or external libraries. I wanted to prove that I understand core web technologies from the ground up, keeping the site lightning-fast and highly maintainable.

For the visuals, I wanted the site to feel deeply authentic to who I am. Blue is my favourite colour, so I used that as the foundation for my personal branding. I designed a modern, “cosmic” dark mode using a deep, immersive blue background (`#0B0D1F`) paired with vibrant accent tones. To make it highly functional, I built a smooth fluid theme toggle using CSS custom properties, allowing users to shift into a clean, crisp off-white light mode layout (`#F7F7FC`) effortlessly.

4.2 Responsive Layout Strategy

Unlike rigid, desktop-only layouts, I prioritised a fully responsive design. By combining CSS Flexbox and Grid with fluid typography math (`clamp()`), the entire layout scales seamlessly. Whether a hiring manager reviews my work on a wide desktop monitor or a smartphone on the go, the content shifts naturally without breaking boundaries or creating awkward horizontal scrolls.

4.3 CV Design via LaTeX

For my CV, I bypassed standard word processors and wrote it from scratch in LaTeX. LaTeX gives me absolute control over typesetting, spacing, and font weight hierarchies. By manually structuring the document layout instead of relying on a crowded pre-made template, I ensured that the file remains perfectly parsable by automated Applicant Tracking Systems (ATS) while staying highly readable and visually satisfying for human recruiters.

5. Tools and Technologies

- **HTML5 & CSS:** Used to construct the entire presentation layer from scratch, utilising CSS variables, grids, and flexboxes to establish a fluid UI.
- **LaTeX:** Selected for drafting a clean, publication-grade résumé, giving me precise control over typography and margins.
- **Python:** The core language used to write my Hangman game, handling the underlying terminal game loops, data processing, and state evaluation.
- **MySQL:** Used to architect my E-sports management database and the backend state storage for the Hangman game, focusing on relational integrity and query optimisation.
- **Git & GitHub:** Used for local version control and remote code hosting, tracking development iterations cleanly through clear commit histories.

6. Strengths and Weaknesses

6.1 Strengths

- **Fluid Responsiveness:** The website adapts instantly to any screen size, preserving layout integrity from mobile devices to desktop monitors.
- **Strong Personal Branding:** The blue-driven cosmic aesthetic stands out immediately, creating a memorable visual theme that reflects my personal style.
- **Clean Code Architecture:** Writing the site without heavy frameworks keeps the codebase lightweight, highly readable, and exceptionally fast to load.
- **ATS-Optimised CV:** The LaTeX résumé strikes a great balance between clean professional design and machine-readable data structures.

6.2 Weaknesses

- **Early-Stage Project Volume:** As an ongoing student, my portfolio currently highlights two main projects, which limits the total breadth of technologies shown.
- **Terminal-Based Gameplay:** The Hangman game executes directly inside the terminal, which demonstrates solid logical skills but lacks immediate visual feedback for non-technical users.

7. Future Improvements

- **Advanced UI Transitions:** I plan to add subtle CSS micro-animations to cards and menu selections to make navigation feel even more dynamic and responsive.
- **Expanding the Project Suite:** As I move forward in my computer science degree, I plan to integrate more complex projects, specifically focusing on full-stack web applications and interactive algorithms.
- **Upgrading to a Graphical Interface:** I would love to refactor the Python Hangman game by replacing the terminal display with a custom GUI (using libraries like Pygame or Tkinter) to give it more visual impact.
- **Integrated Contact Form:** Replacing the basic email links with a built-in contact form would allow recruiters to message me directly without leaving the website.